INCH-POUND

A-A-59340
November 6, 1998

COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, 10,000 POUND CAPACITY AT 48 INCH LOAD CENTER

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description covers the general requirements for a diesel engine driven, front wheel drive, rear wheel steer, pneumatic tired, 10,000 pound at 48 inch load center capacity, forklift truck. The forklift is intended for handling cargo on the 463L pallet and miscellaneous cargo in and around aircraft and warehouses, loading platforms, and docks; on paved or other hard surfaces.

2. SALIENT CHARACTERISTICS.

2.1 Safety. The forklift shall comply with ANSI B56.1, UL 558, and OSHA standards in effect at the time of manufacture. A seatbelt conforming to SAE J 386, and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tipover shall be provided.

2.2 Environmental requirements. The engine shall start without preheating within 5 minutes, and the truck shall operate as specified herein within 15 minutes after engine start in any ambient temperature from 0 to 125 degrees F. When specified, the vehicle shall be winterized to –25 degrees F. As a minimum, a power plant heater, coolant heater, engine oil pan heater, battery heater(s), and brake system air dryer shall be furnished. A 25 foot cable, of sufficient capacity to operate all heaters operating simultaneously, shall be provided.

2.3 Engine. The forklift shall be powered by a standard, commercial, diesel engine. The engine shall operate on diesel fuel in accordance with A-A-52557 without detrimental effects on the engine, or its performance.

2.4 Fuel system. The forklift shall have all necessary filters, water separators, and components required for diesel fuel operation. The forklift shall have a fuel tank of sufficient capacity to allow at least eight hours of continuous operation.

2.5 Starting system. The starter motor shall not be capable of energizing while the engine is running. The starting motor shall energize only when the transmission is in neutral.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to WR-ALC/LEG, 225 Ocmulgee Court, Robins AFB GA 31098-1647.

FSC 3930

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2.6 Transmission. A torque converter and a powershift type of transmission shall be furnished, with at least two speeds forward and two speeds reverse. The transmission shall provide for positive inching control, or declutch control, that permits lifting of rated load at maximum lifting speed while the transmission control is in either forward or reverse direction, without vehicle movement. This control shall also apply the service brakes.

2.7 Hydraulic system. A pressure relief protection device is required along with pump(s), cylinders, control valves, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system. The hydraulic lift system shall be designed so that in the event of hydraulic hose or pump failure, the load shall not descend (freefall) at an uncontrolled rate; maximum lowering speed shall not exceed 40 fpm under any load condition. The hydraulic system shall also be designed to prevent forward tilt in the event of system failure; tilt angle shall not deviate more than two degrees if failure occurs at any position during load handling operations. A manual override for the tilt function shall also be provided to allow for load removal.

2.8 Uprights and carriage. With no load on the forks, the seated operator with seatbelt engaged, shall be able to see at least one fork tip at all lift heights and fork positions.

2.9 Forks and carriage. Fork tines shall have the following dimensions. The forks shall be mounted so that when fully lowered and forklift operating in reverse, there is no damage or degradation to the fork tines, there attachment, or any other component of the carriage, including fork positioner and side shift.

a. Length: 72 ± 1 inches
b. Width: 8 inches (maximum)
c. Thickness: 2.5 inches maximum, except bend in fork heel may exceed 2.5 inches.
d. Fork tip thickness: ½ inch (maximum)
e. Fork taper (underside of forks): Taper from fork tip rearward shall be 30 inches (minimum).
f. Fork carrier assembly width shall not exceed the overall width of the forklift truck.

2.10 Fork positioner. A hydraulically powered fork positioner shall be furnished and controlled by the operator while seated. The fork positioner shall allow tine spacing of 18 inches (or less) to 60 inches (or more), measured between centerlines of the forks.

2.11 Side shift. Hydraulically powered side shift capability shall be furnished. It shall be controlled by the seated operator, and shall provide at least six inch side shift capability (each side of center or 12.0 inches minimum total travel).

2.12 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical rear load guard of at least 48 inches high, measured from the load carrying surfaces of the forks.

2.13 Steering. Power steering, with emergency steering in the event of power failure, shall be furnished.

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2.14 Service brakes. Power brakes conforming to ASME/ANSI B56.1 shall be furnished, except maximum brake pedal force during stopping shall not exceed 80 pounds.

2.15 Parking brake. Parking brake shall conform to ASME/ANSI B56.1.

2.16 Electrical system. A 12 volt direct current (VDC) electrical system shall be furnished. The electrical system shall consist of all electrical components necessary for operation of the forklift. The forklift shall have sufficient electrical grounding to prevent static discharge.

2.17 Instruments and controls. In addition to the instruments supplied on the standard commercial forklift, an hour meter and tachometer shall be installed in the instrument panel. All instruments, except the hourmeter shall illuminate. All load motion controls shall be right hand operation controls and of the self-centering type, i.e., controls shall return to the neutral position when released.

2.18 Lighting. Minimum four sealed beam floodlights, two in front facing forward and two mounted in the rear facing rearward, shall be furnished. An additional floodlight shall be furnished, that is adjustable from inside the cab by the operator while seated; the floodlight shall provide at least 270° rotation capability. Tail light(s) and brake stop light(s) shall be installed. All lights shall be protected by location or guards. Individual operator controlled switches shall be provided for front floodlights, rear floodlights, and adjustable lights.

2.19 Horn. Horn button shall be mounted in the center of the steering wheel.

2.20 Operator's overhead guard. The guard shall be in accordance with ASME/ANSI B56.1. It shall also be a falling-object-protective-structure (FOPS) conforming to ASME/ANSI B56.1. Overhead guard height from ground to top of guard shall be 102 inches maximum.

2.21 Cab. The forklift shall be equipped with an enclosed cab that shall incorporate overhead protection requirements of ASME/ANSI B56.1. The cab shall have a door on each side of the vehicle with at least one window per door which can be opened for ventilation; the windows shall be capable of being locked in either the open or closed positions. The doors shall be provided with hinges that can be serviced without door removal, and shall allow adjustments for door fit. The doors shall be capable of being locked in either the opened or closed positions. In addition to the door windows, the cab shall be furnished with a front windshield, rear window, and roof window. The windshield and all windows shall be safety glass. The cab interior shall be furnished with water resistant non-absorbent thermal insulation.

2.22 Heater and defroster. The manufacturer’s heaviest duty heater and defroster shall be furnished.

2.23 Windshield wipers. There shall be at least three wipers furnished, one on the front windshield, one on the rear window, and one on the roof window.

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2.24 Performance.

a. Capability to safely handle a 10,000 pound, 96 inch per side cube with the center of gravity at the center of the cube, at all lift heights.

b. Lifting speed. At least 55 feet per minute.

c. Lowering speed. Not more than 80 feet per minute with rated load on the forks. Not less than 40 feet per minute with rated load and with engine at idle speed.

d. Right angle turn. With rated load on the forks and forklift positioned perpendicular to a wall, and with the front of the load against the wall, the truck shall be able to back up and make a complete right angle turn (ending up parallel to the wall) within 210 inches.

e. Travel speed. 12 miles per hour (mph) minimum (not to exceed 18 mph), in both forward and reverse, while carrying rated load on the forks. The vehicle speed shall be governed so that maximum speed for safe operations is not exceeded.

f. Slope ascension. With the truck facing up the slope and with rated load on the forks, it shall be able to accelerate up a 20 percent slope from a dead stop.

g. Upright tilt. With no load on the truck, the forks shall have at least 3.0 degrees forward tilt, and at least 6.0 degrees of rear tilt.

h. Upright height with forks on ground (collapsed mast height): 130 inches maximum, with no load on the forks.

i. Maximum fork lift height. At least 168 inches with rated load on forks and forks horizontal.

j. Ground clearance. With rated load on the forks and mast vertical, ground clearance beneath the mast shall be at least 5.0 inches. With no load on the forks, the ground clearance of the truck, other than the mast and axles, shall be at least 7.0 inches.

k. Load drift. With hydraulic fluid at normal operating temperature, lift assembly shall hold rated load at maximum lift height for at least 2.0 minutes with not more than 1-3/4 inch of vertical drift, and not more than 1.0 degree of rotational drift.

l. Stability. The forklift shall meet the ASME/ANSI B56.1 "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements.

m. Noise limits. Maximum allowable noise level shall not exceed 84dB(A) at the operator's ear. If the noise level at the operator's ear exceeds 84 dB(A) (85 to 92 dB(A)), a permanent warning decal or placard shall be installed inside the cab clearly visible to the operator. Noise level limits that exceed 92dB(A) are not acceptable.

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2.25 Painting. The forklift shall be primed and painted using standard commercial practices. Unless otherwise specified, color shall be 24052 of FED-STD-595.

2.26 Markings, data plates, and instruction plates. Corrosion resistant identification plate, hydraulic schematic, wiring schematic, and all warning and instruction plates shall be provided and mounted on the vehicle. The identification plate shall contain the following information:

Truck, Lift, Fork
Capacity: 10,000 pounds at 48 inch load center
Gross Vehicle Weight (GVW):
National Stock Number (NSN):
Contract Number:
Serial Number:
Model Number:
Registration Number:
Manufactured by:
    Name:
    Address:
    Service Phone Number:
US Property:

2.27 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip coating compound, or be furnished with a nonslip metal or treadplate surface.

2.28 Lifting and tiedown provisions. There shall be permanently installed lifting devices that enable the forklift to be lifted in its normal travel configuration, and tiedown devices that enable the forklift to be tied down to the floor of the transportation media. All devices shall have an inside diameter of 3.0 inches and minimum structural safety factor of 2:1, based on static load. The notation “LIFT HERE” and “TIEDOWN” shall be stenciled, in black, near each lifting and tiedown device.

2.29 Electromagnetic interference (EMI). The forklift shall comply with the EMI requirements of SAE J551.

2.30 Highway transportability. The truck, when loaded on a semitrailer, shall be within the highway permit limits for all states.

3. REGULATORY REQUIREMENTS.

3.1 The contractor is encouraged to use recovered materials to the maximum extent possible. However, used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the forklift. The forklift shall comply with applicable Environmental Protection Agency (EPA) regulations.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this commercial item description and conforms to the producer's own drawings, specifications, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and therefore as may be otherwise provided for under the provisions of the contract.

5. Preservation, packaging, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from the receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities, labeling and marking shall be as specified in the contract and/or order.

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